

Problem Set 3

First Order Linear Differential Equations

PSet Tips

- ◆ Remember to explain the reasoning behind all of your answers.
- ◆ Keep your work organized and assign a page to each problem on Gradescope.
- ◆ You don't have to do this by yourself — office hours, the MQC, and your fellow students are great resources if you get stuck.

In problems 1 and 2, find the general solution to the differential equations.

1 $(x^2 + 1) \frac{dy}{dx} + 2xy - 3x = 0$

2 $xy' + y = 4\sqrt{x}; \quad x > 0.$

In problems 3–7, solve the initial value problems.

3 $\frac{dy}{dx} = -2xy + 2x e^{(-x^2)}; \quad y(0) = 2.$

4 $xy' = 2y + x^3 \sin(x); \quad y(1) = 0.$

5 $(x^2 + 1)y' + 5xy = x; \quad y(0) = 2.$

6 $xy' + 4y = \frac{e^{-x}}{x^3}; \quad y(1) = 0, \ x > 0.$

7 $(1+x)y' + y = \cos(x); \quad y(0) = 1, \ 1+x > 0.$